

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce

Department of Information Technology (B.Sc.I.T Semester III)

Data structure

Practical – 4

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Queue Application:

a. Write a program to: a. Implement a queue using an array.

Code:-

```
from array import array
queue = array("i")
queue.append(10)
queue.append(20)
queue.append(30)
print("Queue:", queue)
queue.pop(0)
print("After Dequeue:", queue)
print("S056 Shweta Singh")
```

Output:-

```
===== RESTART: C:/Users/itlak
Queue: array('i', [10, 20, 30])
After Dequeue: array('i', [20, 30])
S056 Shweta Singh
```

b. Simulate a simple queuing system (e.g., customer service queue).\

Code:-

```
import time
que= [ ("Shweta",2), ("Priya",3), ("Raj",4), ("Seema",6) ]
print("-----Booking-----")
print(que)
while que:
    job, timer = que.pop(0)
    print(f"Running:{job} {timer}sec")
    time.sleep(timer)
print(que)
print("S056 Shweta Singh")
```

Output:-

```
===== RESTART: C:/Users/itlab/Downloads/ab
-----Booking-----
[('Shweta', 2), ('Priya', 3), ('Raj', 4), ('Seema', 6)]
Running:Shweta 2sec
Running:Priya 3sec
Running:Raj 4sec
Running:Seema 6sec
[]
S056 Shweta Singh
```